

Tracing Hawaii's Trades Apprenticeship Pipeline from Enrollment to Exit

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HAWAI'I DATA SCIENCE



Strengthening the CyberInfrastructure Professionals Ecosystem (SCIPLE)

CyberInfrastructure (CI) - Summer Immersion Program (SIP)

SCIPLE CI-SIP Fellow - Award #2417946



CI-PACIFIC
PROFESSIONALS

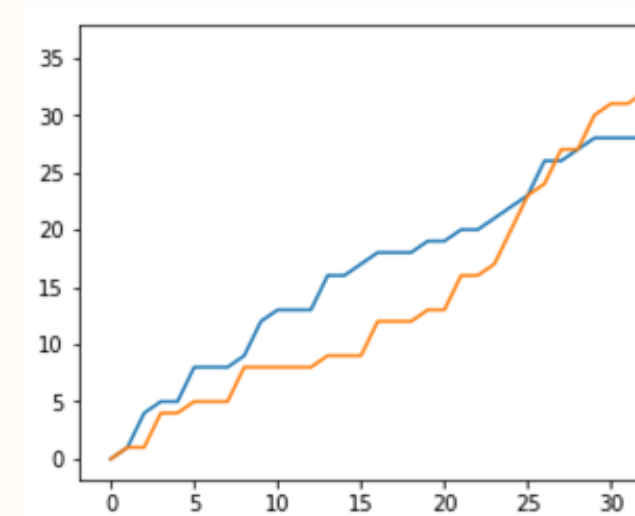
**“In most every business, you learn by doing.”
— Andrew Yang**

- 1. Introduction**
- 2. Data Source**
- 3. Data Process**
- 4. Insights and Analysis**
- 5. Conclusion**
- 6. Acknowledgements**

Introduction.

Background + Education

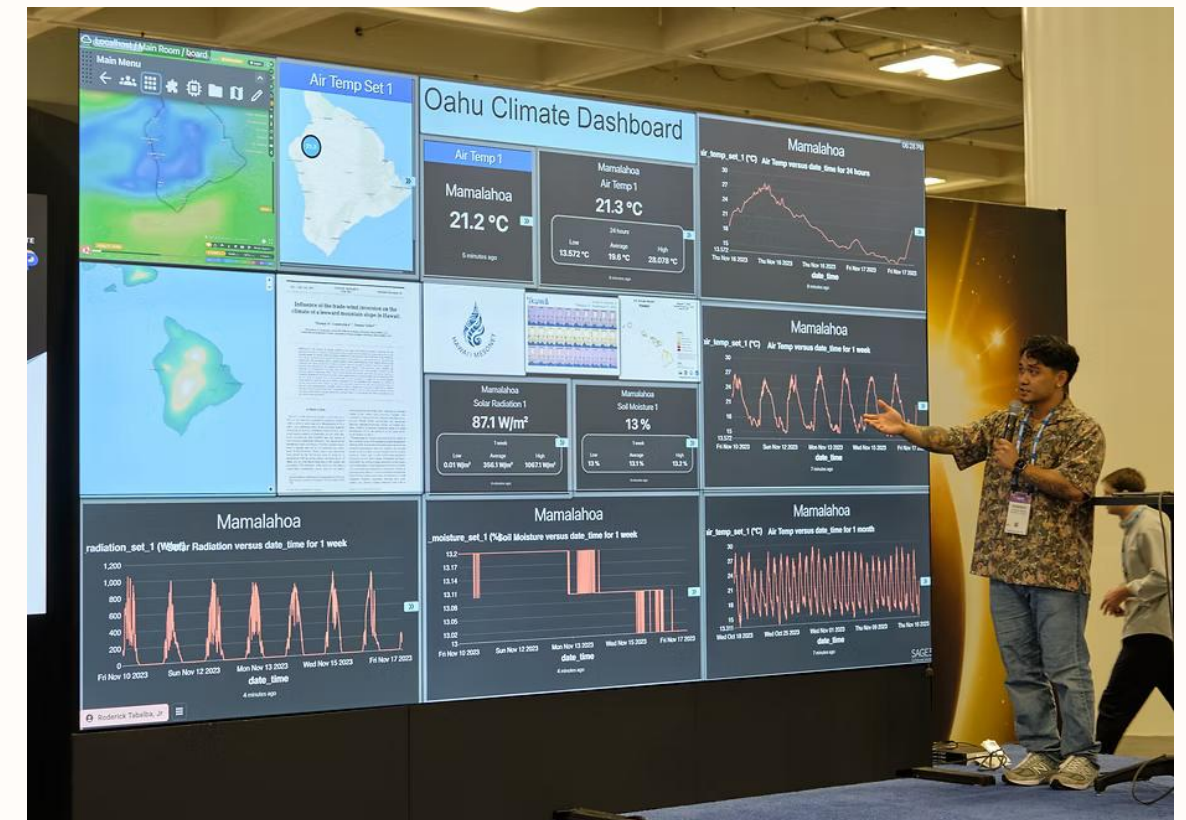
- Born and raised in Honolulu, Hawai‘i
- Quantitative Economics (Econometrics),
Class of 2027
- Taught to examine mechanisms behind
outcomes
- Guided by "data over assumptions"



Research Scope

Why SC�PE CI-SIP?

- Wanted real research experience
- Wanted to apply data science + econ to **real** Hawai‘i economic issues
- Wanted a project with structure, mentorship, and impact



Why I Choose a Project on HI Apprenticeship:

- “Just go into the trades” was common advice I grew up hearing
- Gen Z job anxiety + unclear pathways
- Mentions of skilled trades shortage on the news
- Hawai‘i’s apprenticeship system ≠ national system
- Wanted to see what the data actually shows



Research Question

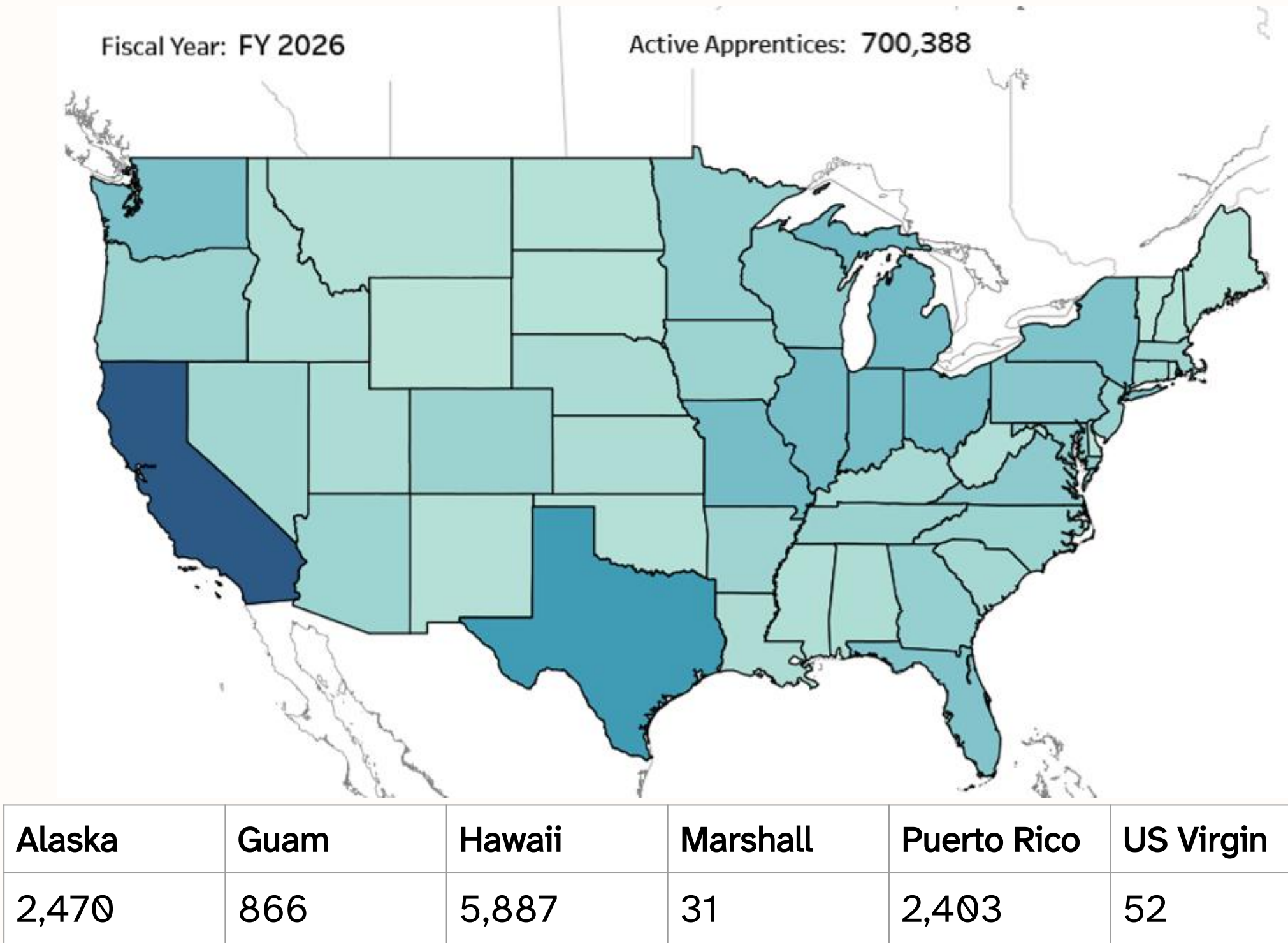
Which journeyworker occupations face the **greatest labor shortages specially in Hawai‘i**, and which apprenticeship programs actually feed those pipelines?



Source: Hawaiian Electric Apprenticeship-program (n.d.)

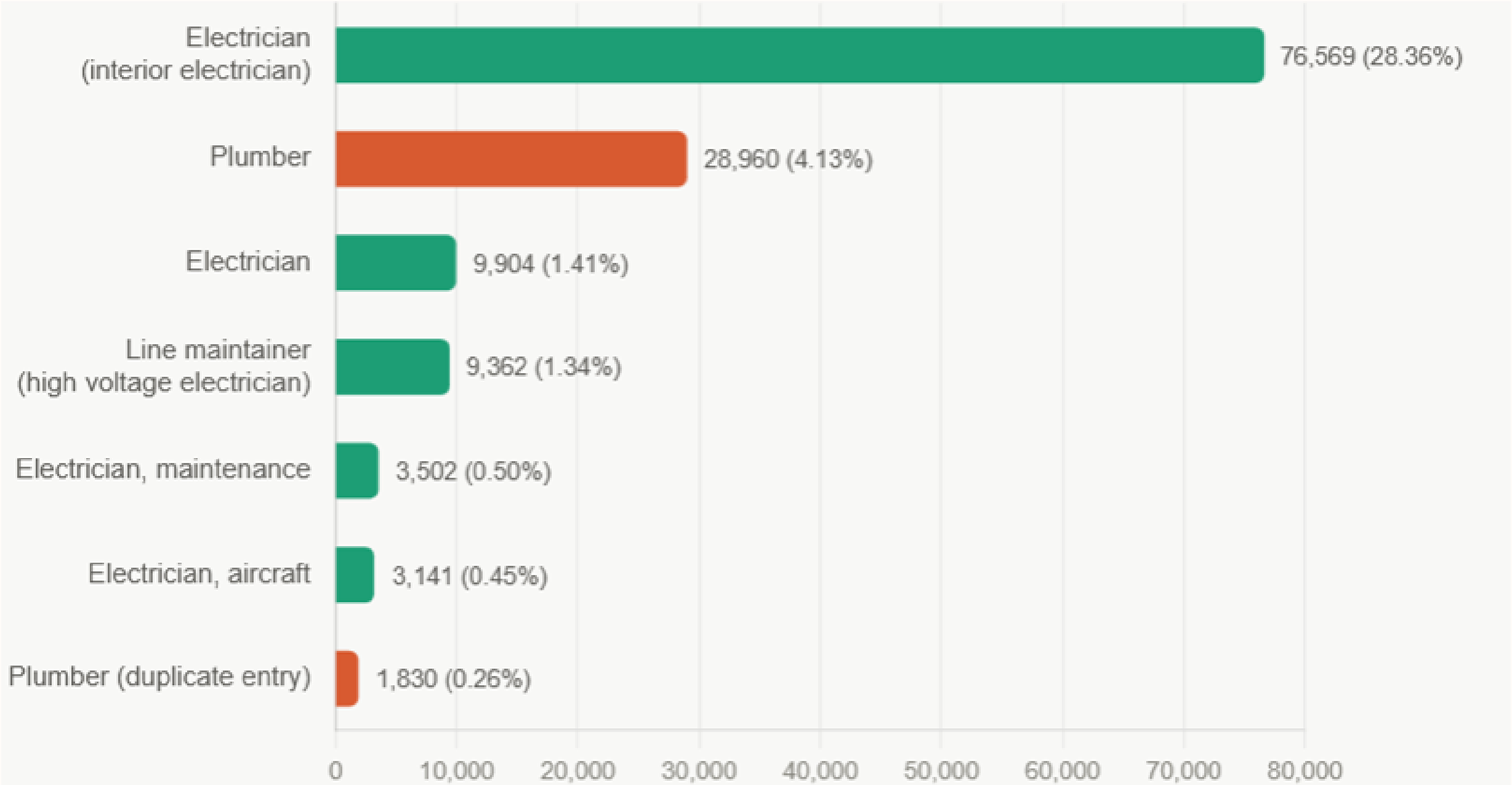
Data Source.

U.S. Department of Labor: Registered Apprenticeship Partners Information Database System



- Official federal system for managing Registered Apprenticeship records
- Tracks programs, occupations, wages, and apprentice progress
- Used by sponsors for registration

Data Challenges and Limitations p.1



Occupation, Broad	Problem	Apprentices, Number
Electrician	Title variants x5	102,478
Plumber	Duplicate entries	30,790

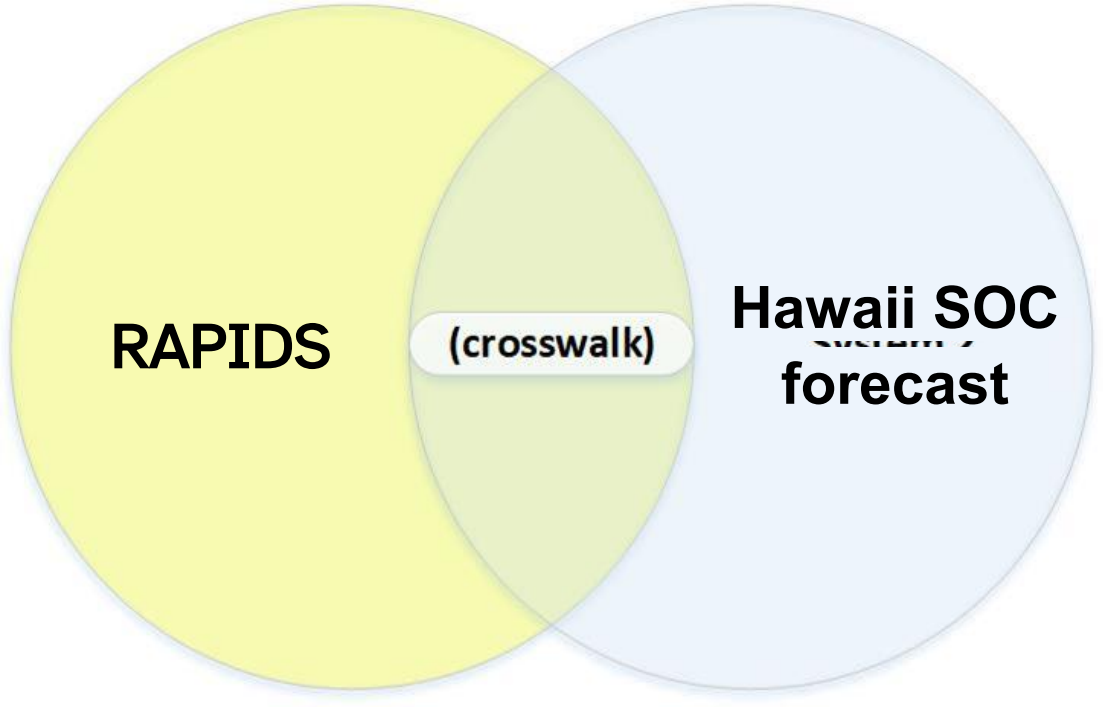
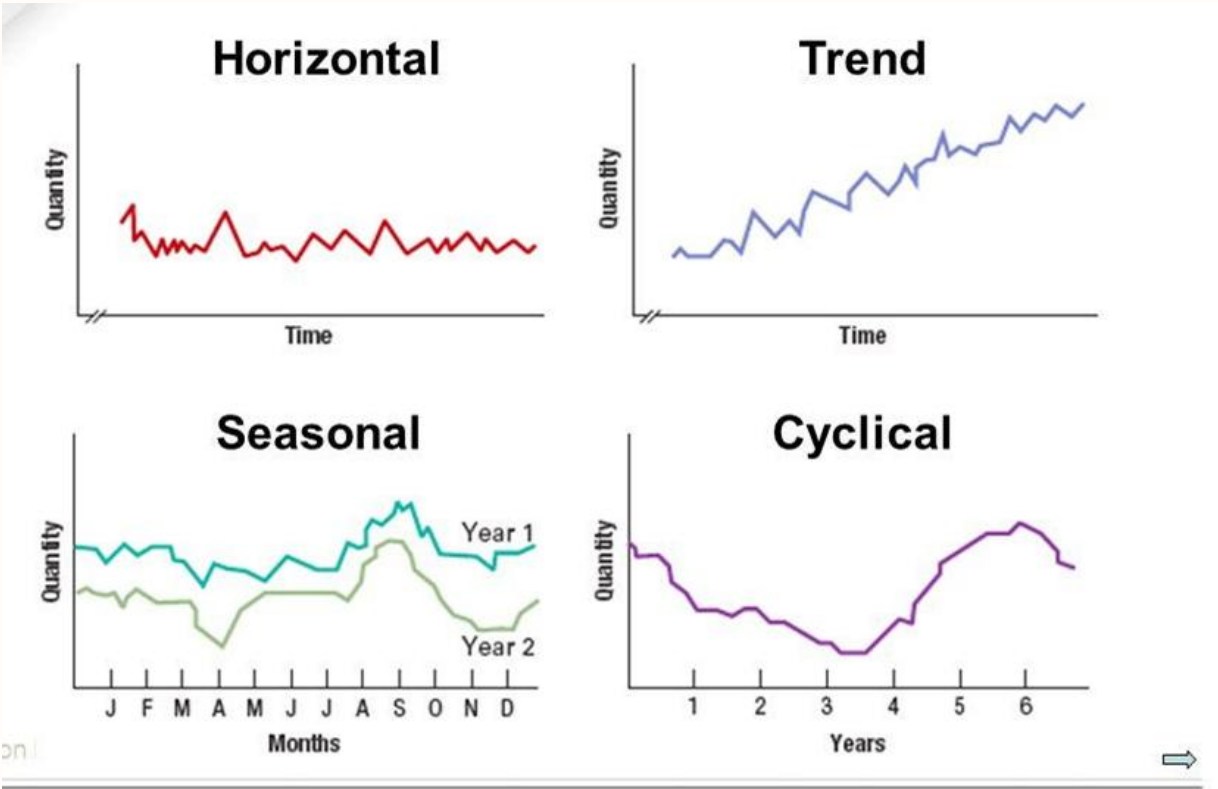
- No standardized occupation titles
- No built-in tools to compare programs or completion times
- Data obtained stopped at 2024

State of Hawaii: FY 2024-2034 Occupation Forecasts

Occupation Title	Employment		Growth (2024-34)		Annual Growth	Annual Average Openings				Education & Training Requirements		
	2024	2034	Net	Percent		Change	Transfer	Exits	Total	Education	Work Exp.	Job Training
Brickmasons and Blockmasons	70	70	0	1.4%	0.1%	0	*	*	10	HS/equiv	None	Apprentice
Carpenters	8,320	8,710	390	4.6%	0.5%	40	360	240	640	HS/equiv	None	Apprentice
Construction Laborers	5,940	6,430	490	8.2%	0.8%	50	290	200	540	None	None	Short OJT
Glaziers	400	410	20	4.8%	0.5%	*	20	10	30	HS/equiv	None	Apprentice

- Annual average openings by broad occupation drives Hawaii labor demand estimate
- 1:1 replacement ratio assumed (exit/transfer → hire)
- Missing data are written as *

Data Challenges and Limitations p.2



- Demand projections use long-term averages
- No direct apprenticeship trade and SOC crosswalk
- Excel format requires cleanup before merging
- Missing data for several occupation codes

Occupation Title	Employment		Growth (2024-34)		Annual Growth	Annual Average Openings				Education & Training Requirements		
	2024	2034	Net	Percent		Change	Transfer	Exits	Total	Education	Work Exp.	Job Training

Data
Process.

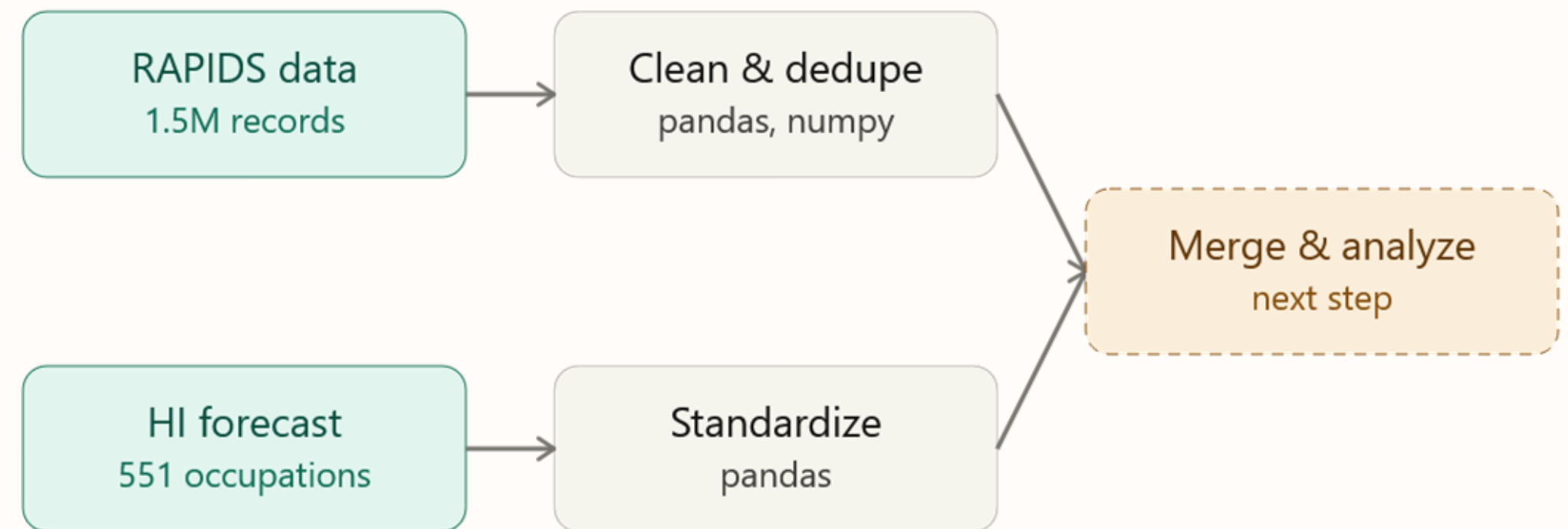
How I did it

IDE: VSCode + Jupyter extension

- Ran on my laptop, no HPC needed

Language & Libraries:

- **Python (pandas, numpy)**
 - Cleaning ~1.5M rows (dedup, dtype fixes, missing-data diagnostics)
- **Parquet**: Compressed Data Export
- **Plotly**: Interactive Data Visualization

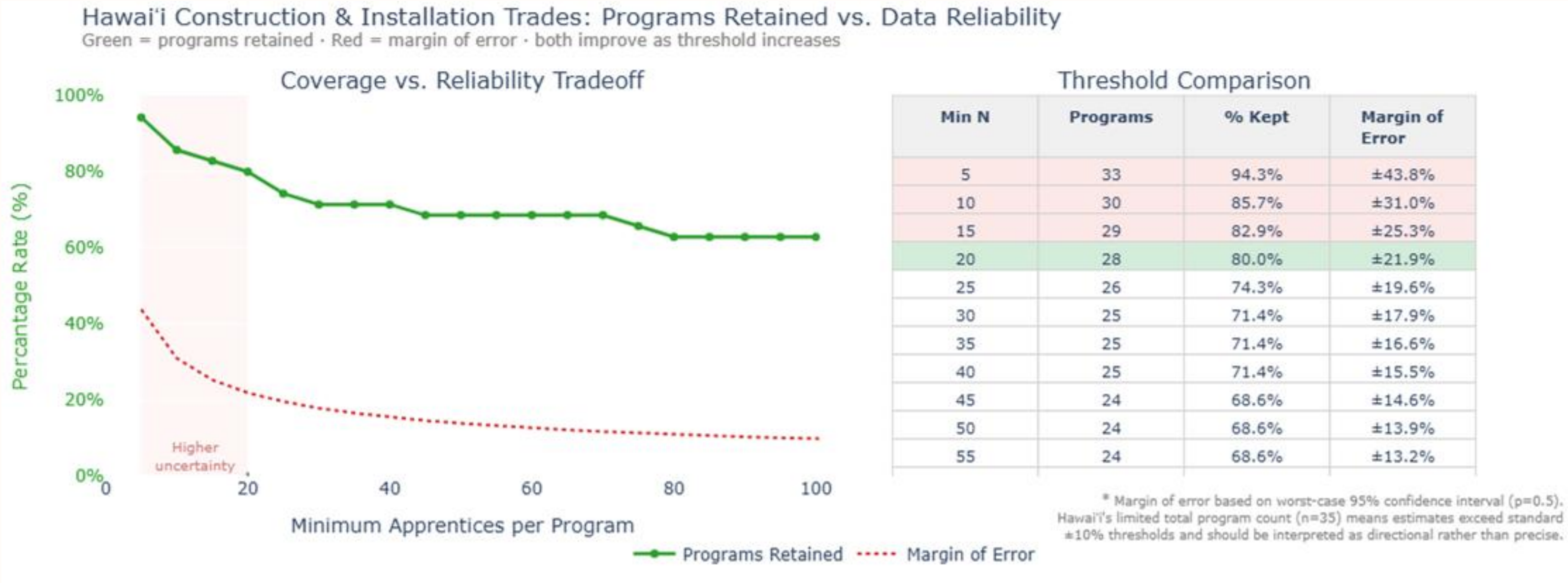


Teal = data source · gray = cleaning step · dashed = planned next step

For code on data cleaning, check out my github:

1. 01_apprenticeships_data_cleaning.ipynb
2. 02_hawaii_forecast_cleaning.ipynb

Statistical Reliability Validation



1. Luna's Tex Mex Cantina

★★★★☆ 3.8 (51 reviews)

📍 Kaka'ako • \$\$ • **Closes in 12 min**

2. La Birria

★★★★★ 5.0 (3 reviews)

📍 Aiea • **Open** until 7:00 PM

Selection Bias on Applicants:

Most Popular Apprenticeship known is the **Pearl Harbor Naval Shipyard**. It also has one of the most exceptional completion rates, all more than 3 standard deviations above national average for the program's trade.

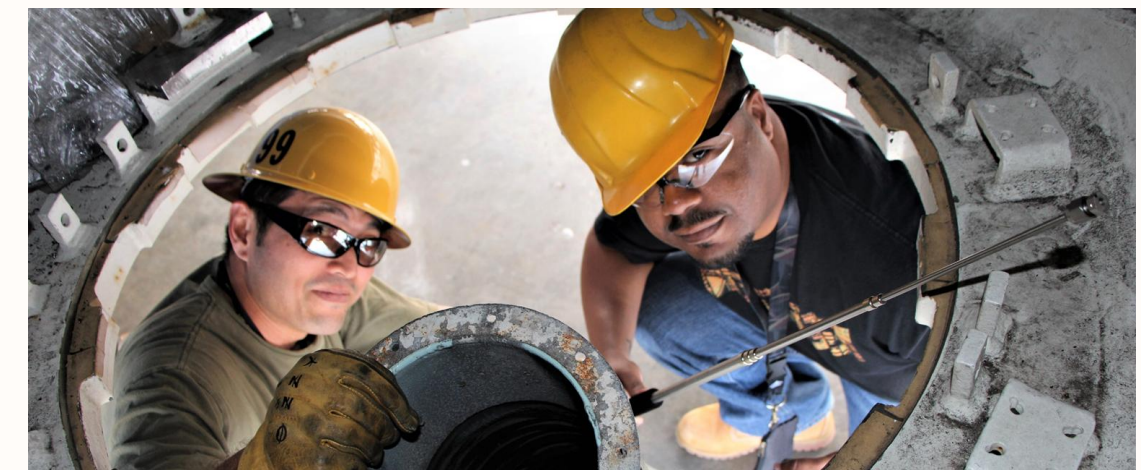
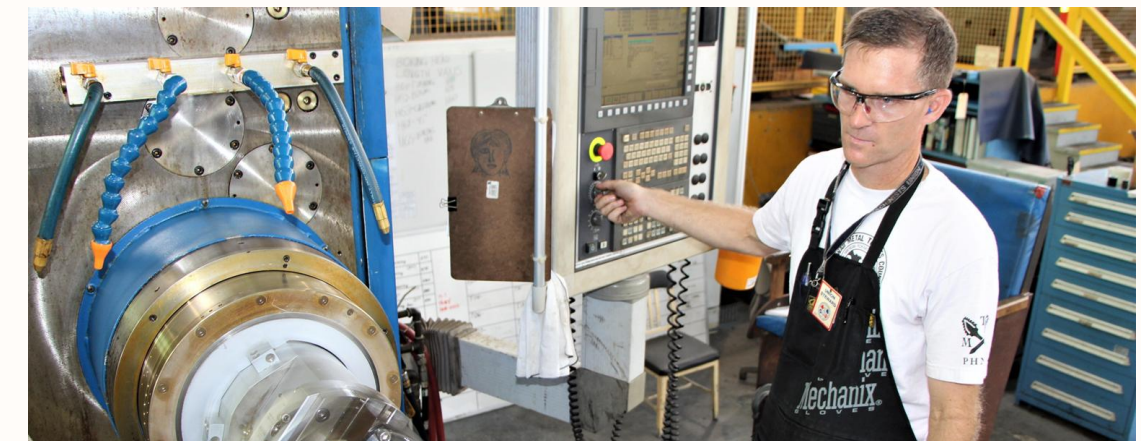
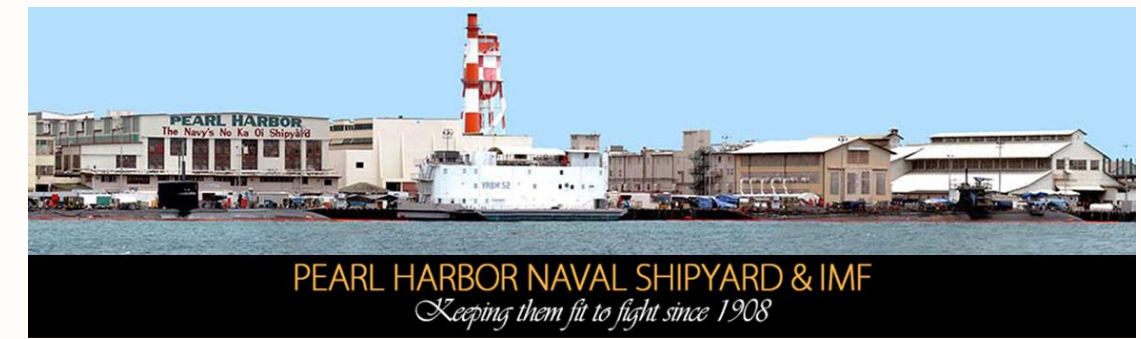
Requirements:

- Requires an Office of Personnel Management (OPM) exam
- Requires U.S. citizenship
- Requires security clearance eligibility
- Requires medical clearance

Other additional Information:

- Assigns apprentices to trades based on testing
- Is historically prestigious and competitive
- Caps applications in some cycles

Source: *Apprentice Program*. (2023). Navy.mil. <https://www.navsea.navy.mil/Home/Shipyards/PHNS-IMF/Careers/Apprenticeship/Apprenticeship-Postings/>



Privacy

- Final data will **NOT be uploaded** due to ease of obtaining information of narrow cohort (you will not see it on GitHub)
- I will look into creating an interactive dashboard website with data built-in

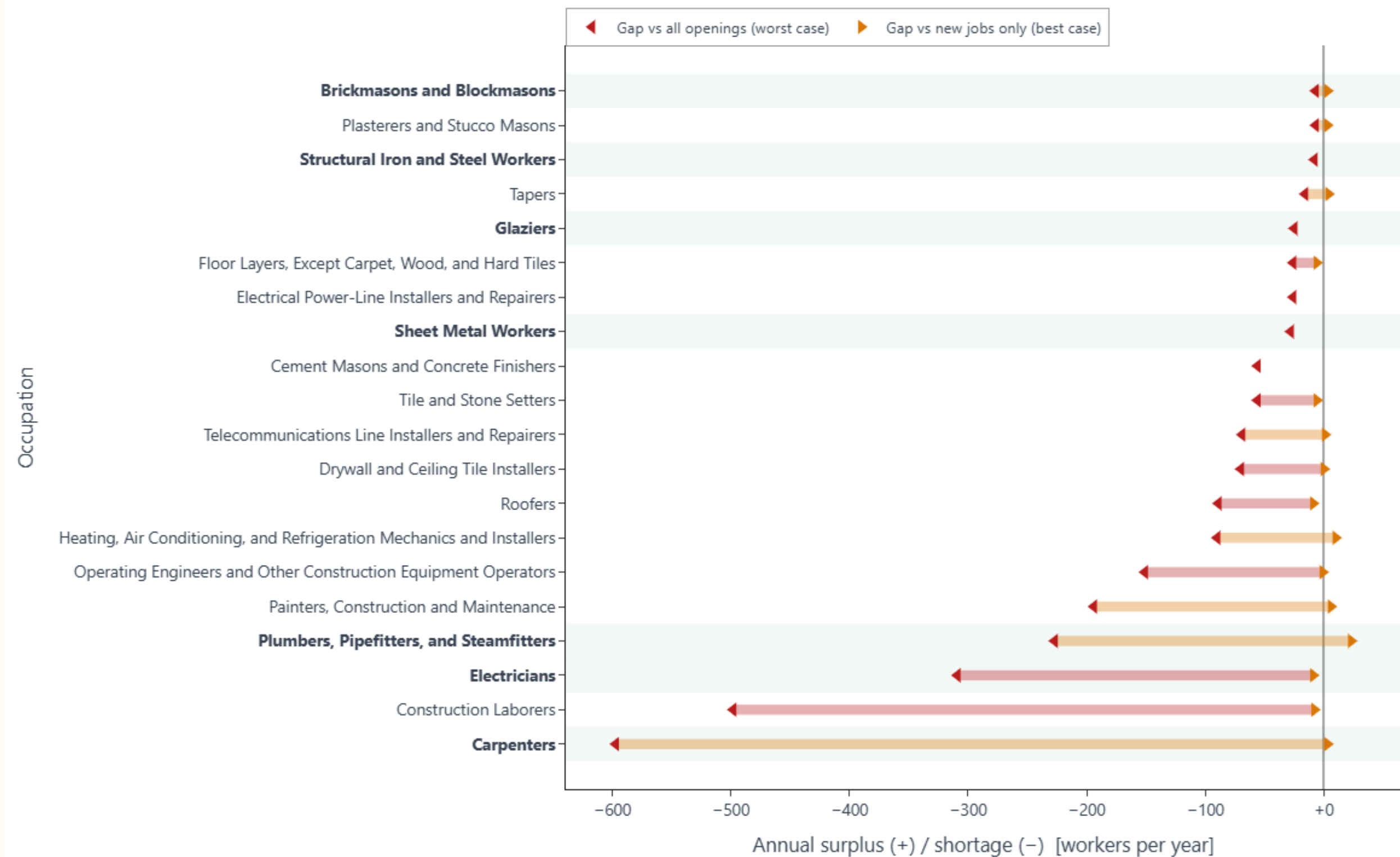


Insights.

Apprenticeship Supply vs Labor Demand (the gap chart)

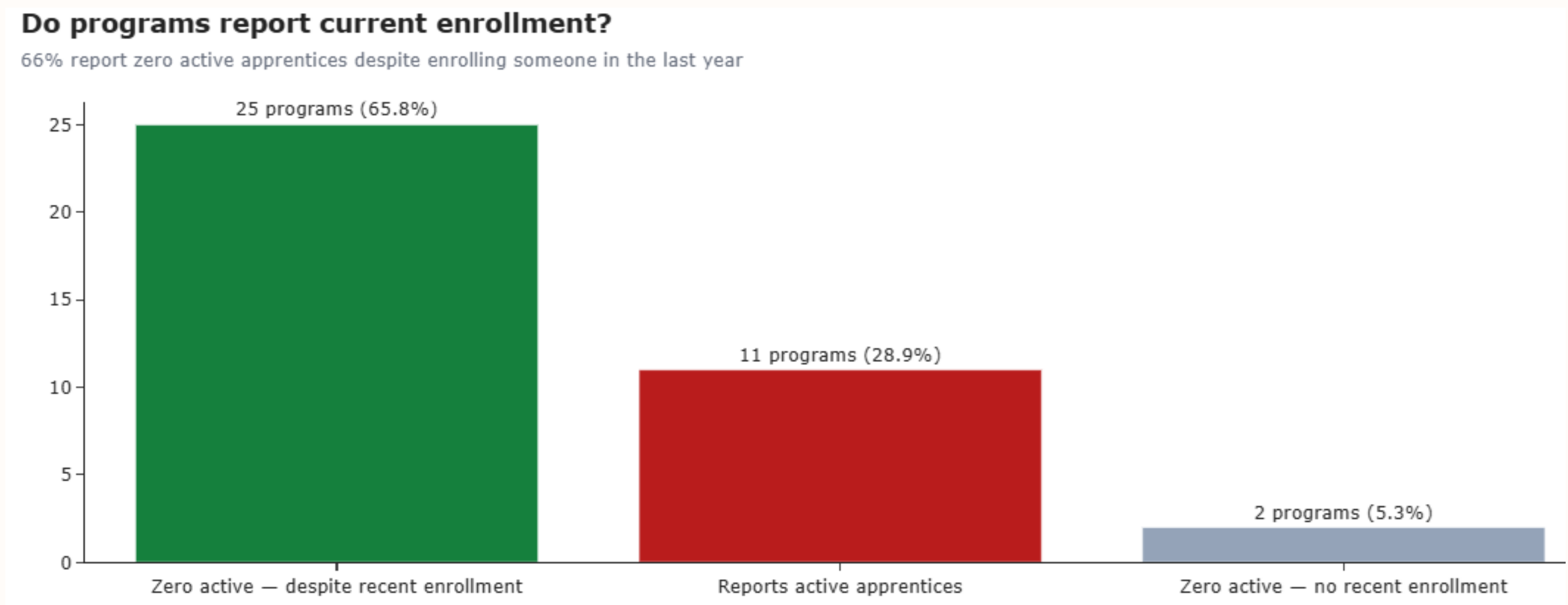
Apprenticeship Supply vs Labor Demand — Hawai'i Construction & Installation Trades

Values = avg annual graduates minus annual demand. Left of zero = shortfall. Range = uncertainty between growth-only vs all-openings demand. Shaded Row = ■ BLS "Apprenticeship" training category



- Shaded rows = Bureau of Statistics training classification under apprenticeship
- ~600 carpenters / year below demand
- Limited resources should target greatest demand of labor first

Active Apprentice Reporting Gap (the enrollment chart)

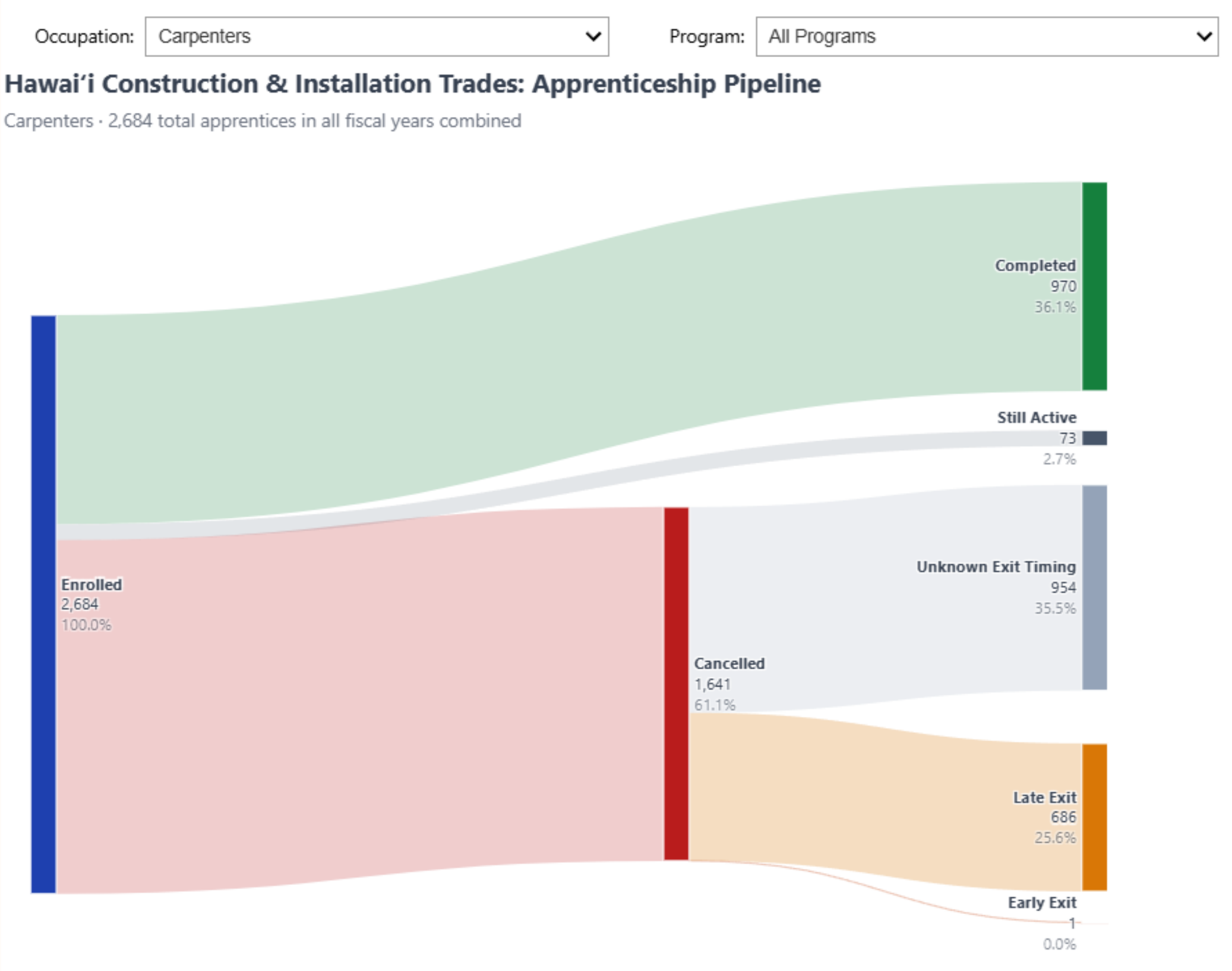


- Active apprentice counts are under-reported across many programs
- Shortage estimates should be read as a floor, not an exact number
- True shortage may be larger or smaller than the headline gap chart shows

	total_apprentices	completion_rate
OCCUPATION_GROUP_NAME		
Carpenters	2684	36.1
Electricians	1378	42.0
Roofers	943	5.4
Plumbers, Pipefitters, and Steamfitters	895	67.4
Painters, Construction and Maintenance	890	41.0
Construction Laborers	723	74.1
Drywall and Ceiling Tile Installers	653	29.4
Tile and Stone Setters	449	7.8
Heating, Air Conditioning, and Refrigeration Mechanics and Installers	358	86.9
Sheet Metal Workers	307	65.5
Reinforcing Iron and Rebar Workers	304	44.7
Cement Masons and Concrete Finishers	264	21.6
Tapers	264	36.0
Floor Layers, Except Carpet, Wood, and Hard Tiles	225	19.6
Glaziers	196	29.1
Electrical Power-Line Installers and Repairers	189	89.9
Plasterers and Stucco Masons	179	16.2
Elevator and Escalator Installers and Repairers	172	65.1
Operating Engineers and Other Construction Equipment Operators	167	52.7
Structural Iron and Steel Workers	144	46.5
Brickmasons and Blockmasons	142	20.4
Insulation Workers, Floor, Ceiling, and Wall	117	67.5
Paving, Surfacing, and Tamping Equipment Operators	42	73.8
Millwrights	34	67.6
Stonemasons	20	5.0
Boilermakers	13	76.9

- **Completion rates rarely get reported nor discussed**
- **Raises the question: “How do we improve programs beyond just recruiting more?”**
- **Full occupation breakdowns included here for reference**

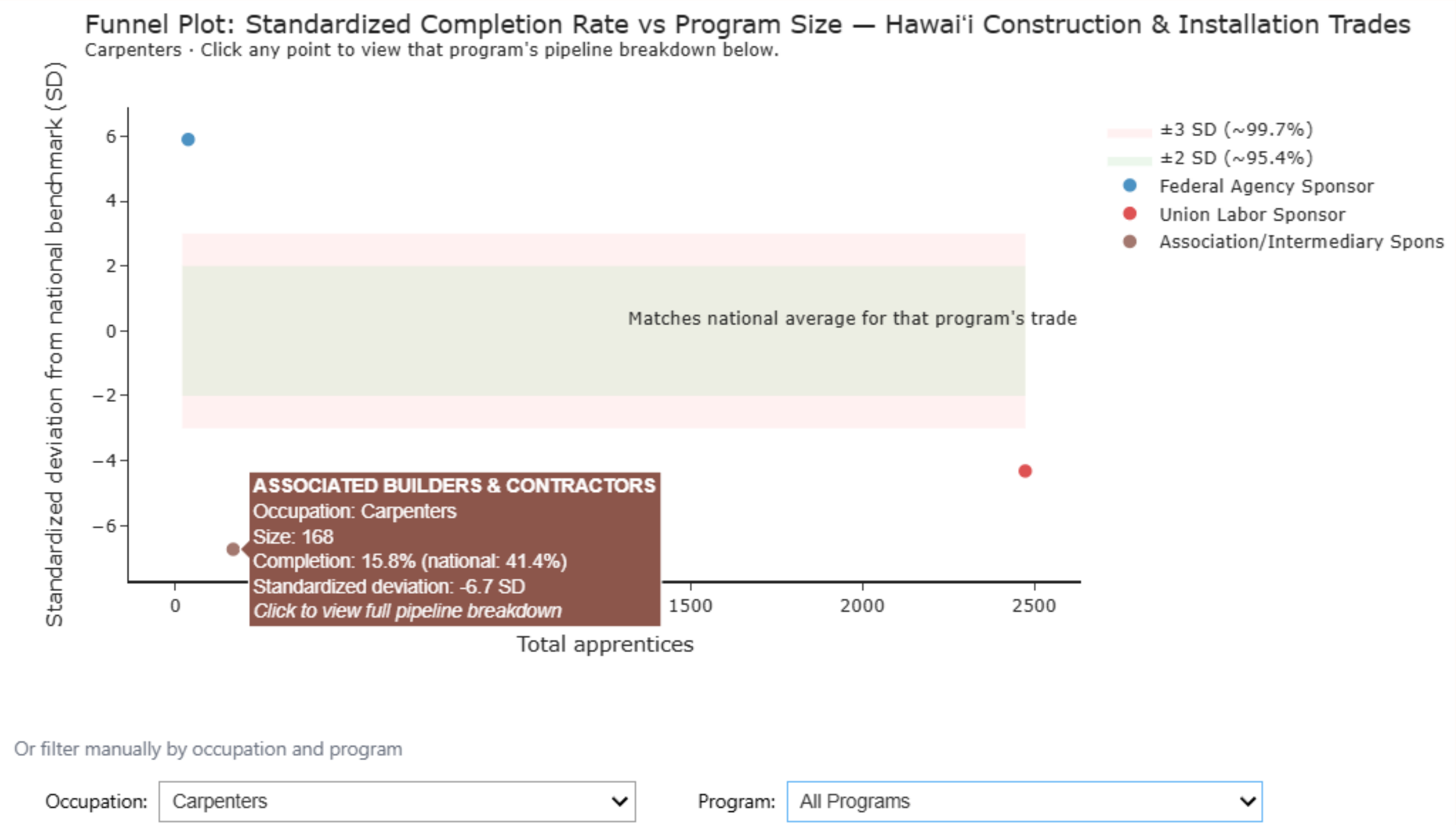
Program Dashboard Illustration



Sankey Diagram:

- Shows for all completion and cancellation rates
- Missing in Reporting: Active Apprentices
- Ability to filter by occupation, and then by program
- Shown as cumulative totals, not a live count

Program Illustration Part 2



- Funnel Plot:
- Compares programs to the national completion average
 - Diagnostic, not a "good/bad" verdict
 - Watch for selection bias

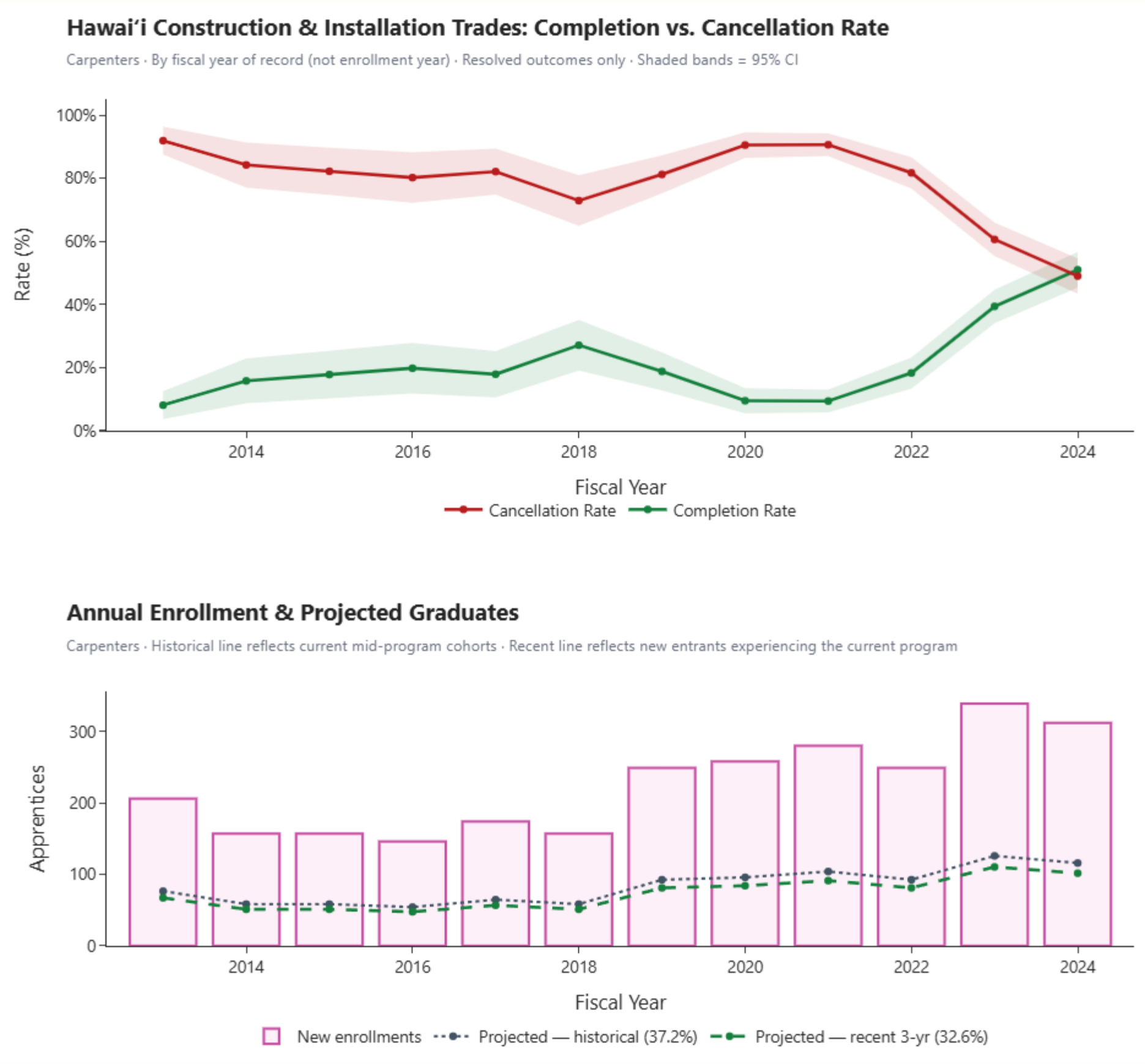
Program Illustration Part 3

Completion vs. Cancellation:

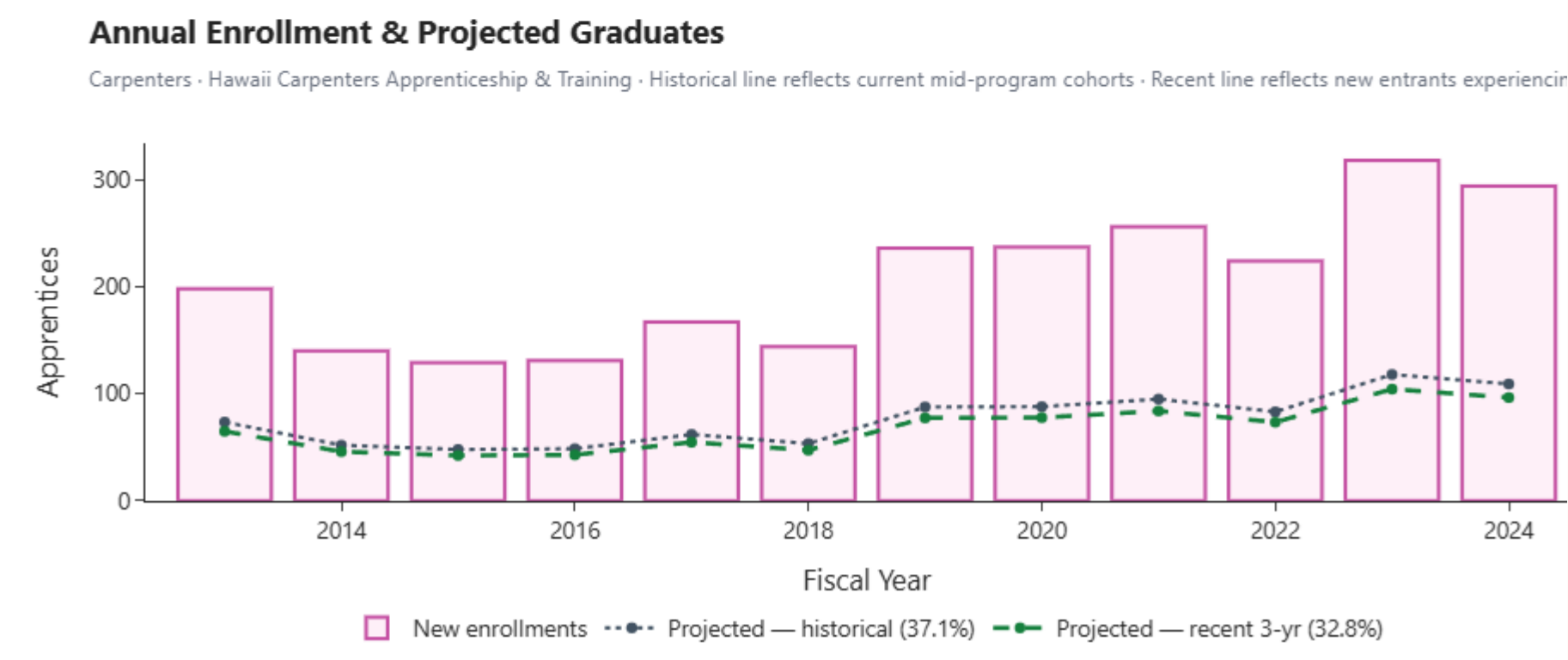
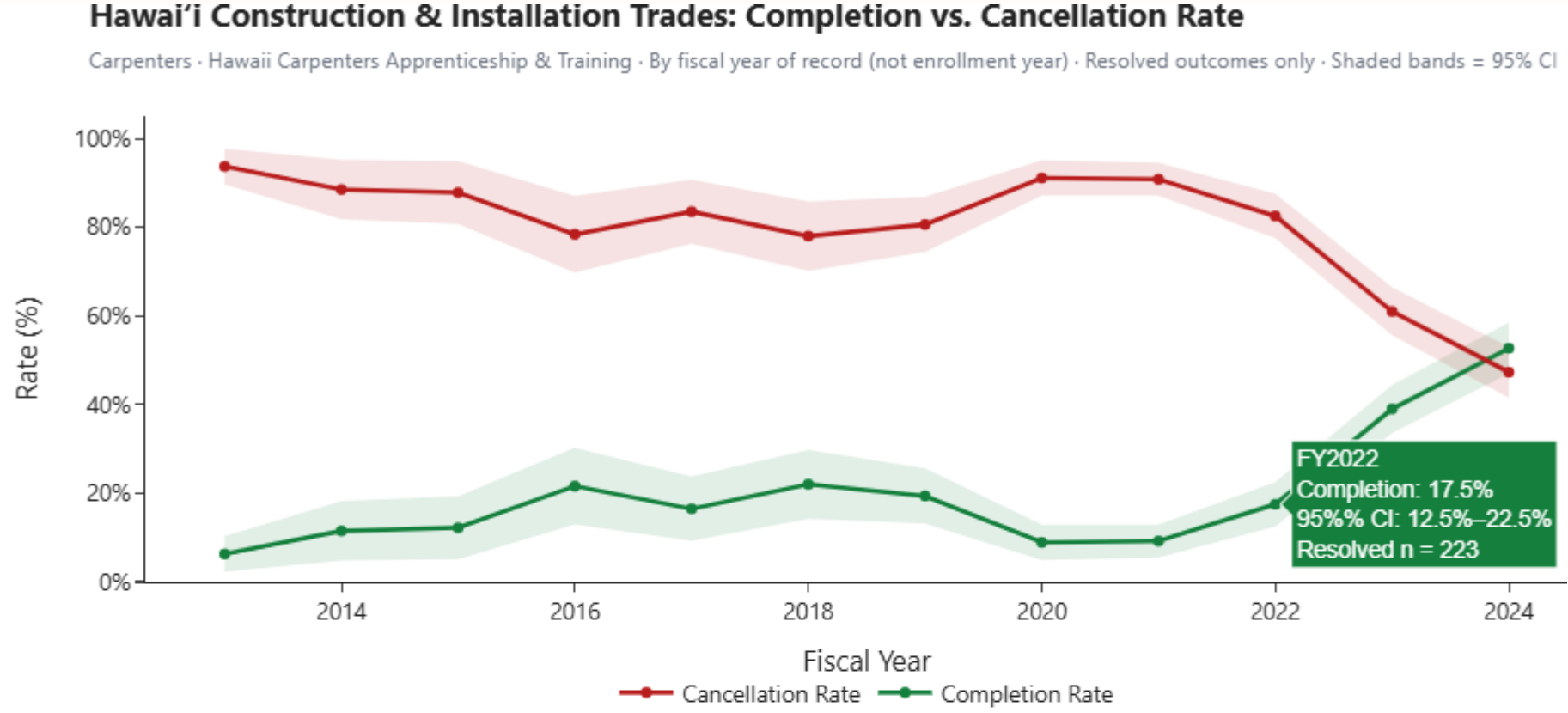
- Reported Completion vs. Cancellation Rate by Fiscal Year
- Caution: Enrolled cohorts $\neq$ graduate cohorts

Annual Enrollment:

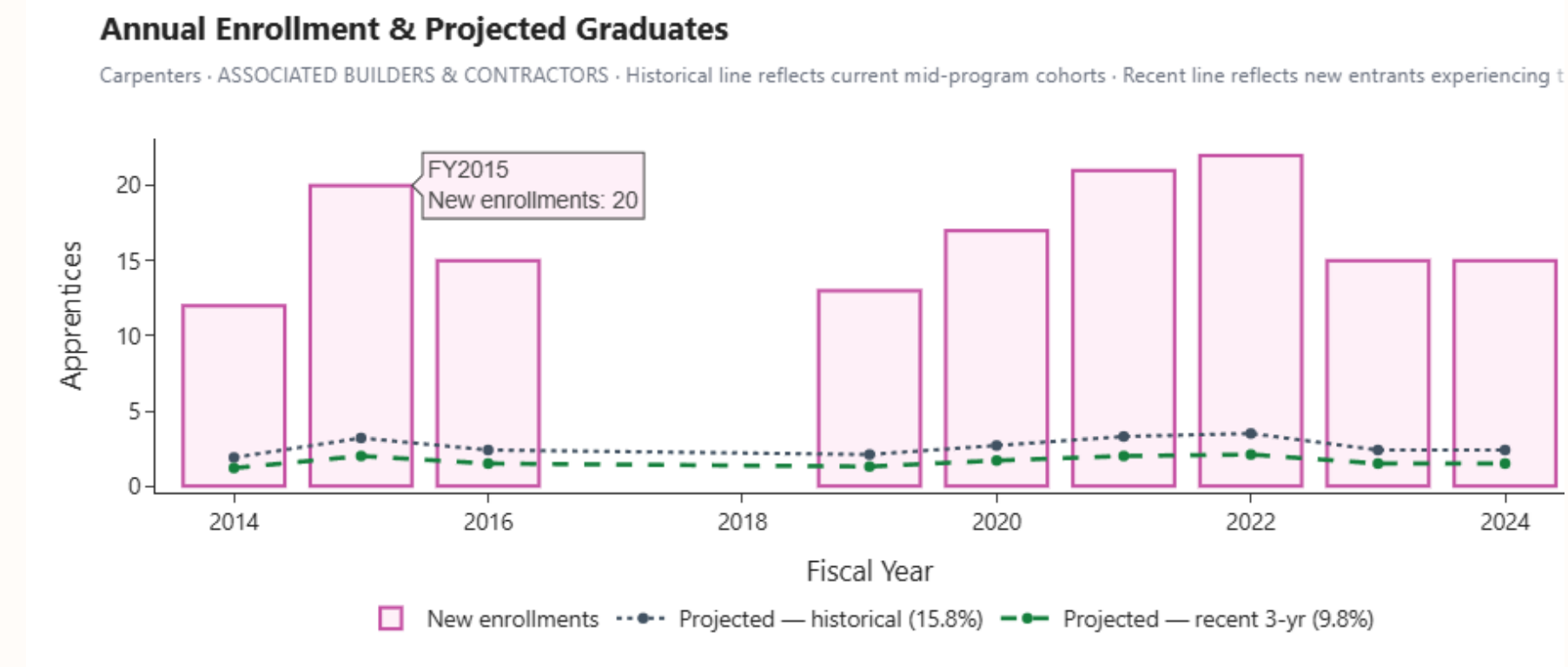
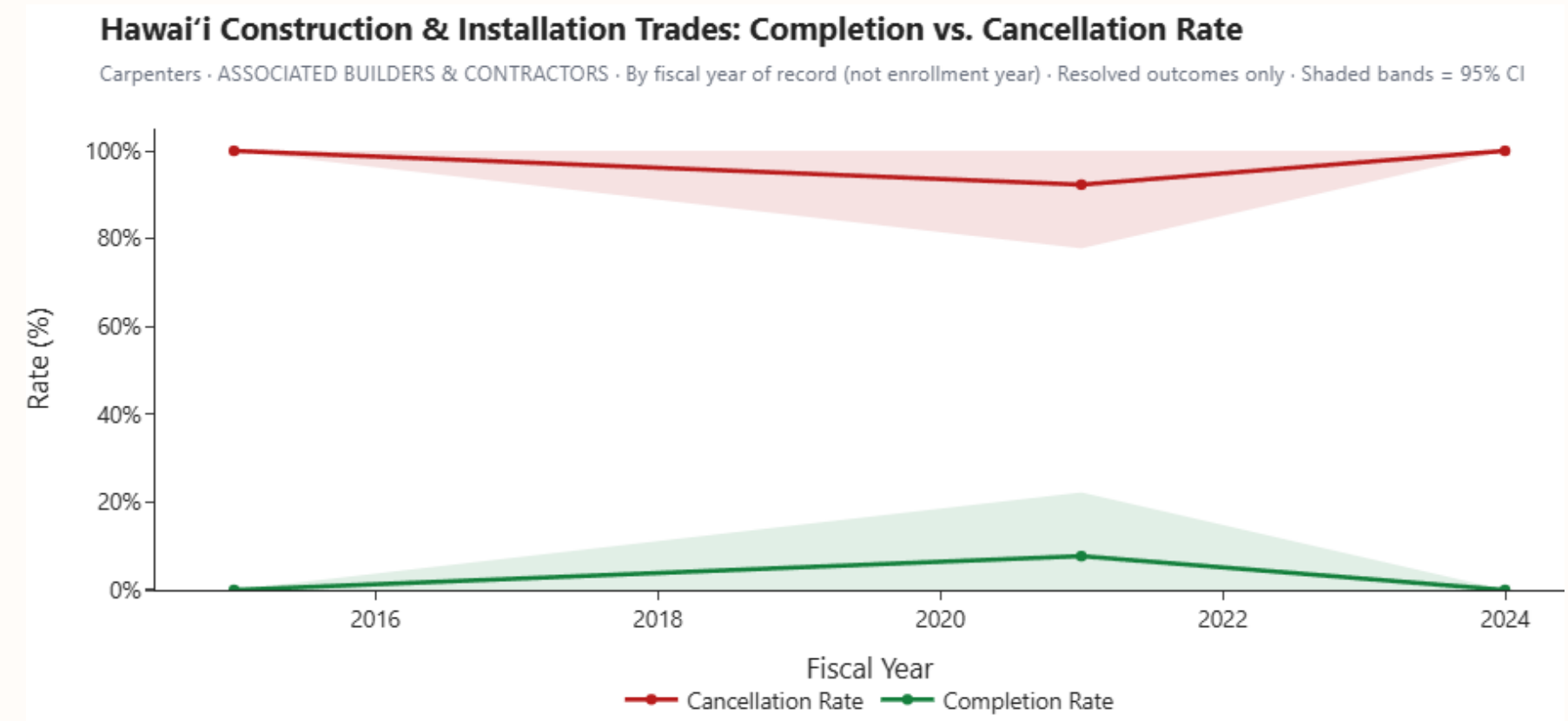
- Reports number of new enrollments
- Worth looking into enrollment from active recruitment success



Program Comparison of Carpenter Occupation



Hawaii Carpenters Apprenticeship & Training

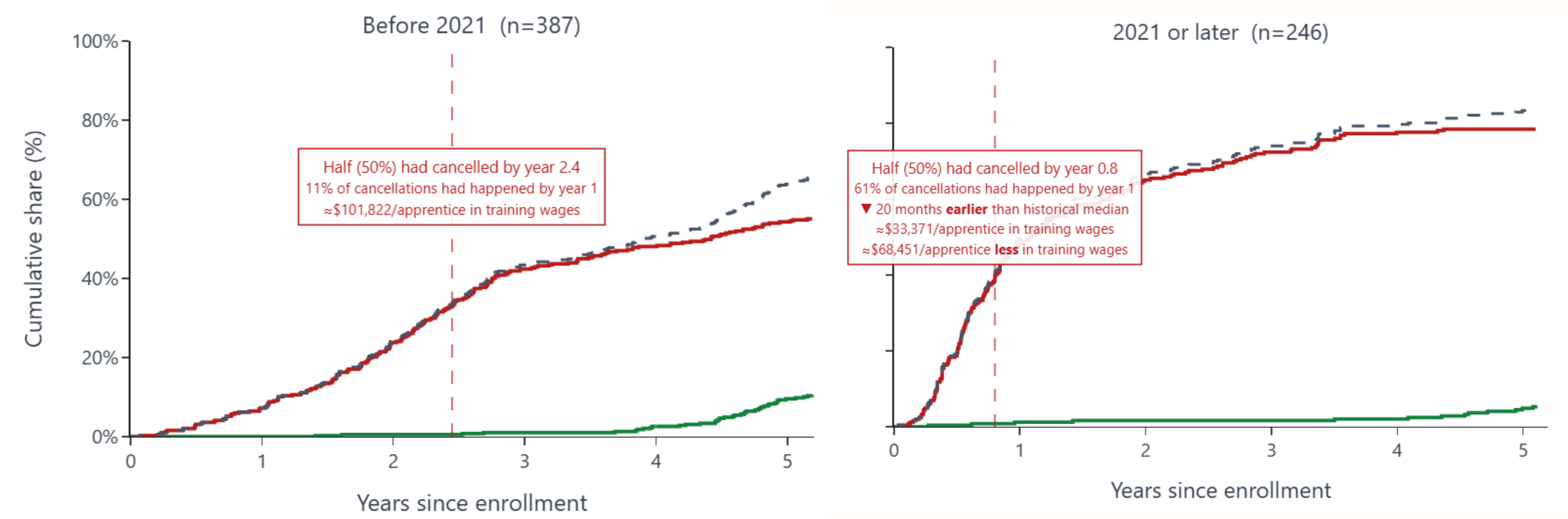


Associated Builders & Contractors (ABC)

Program Illustration Part 4

When do apprentices leave or finish?

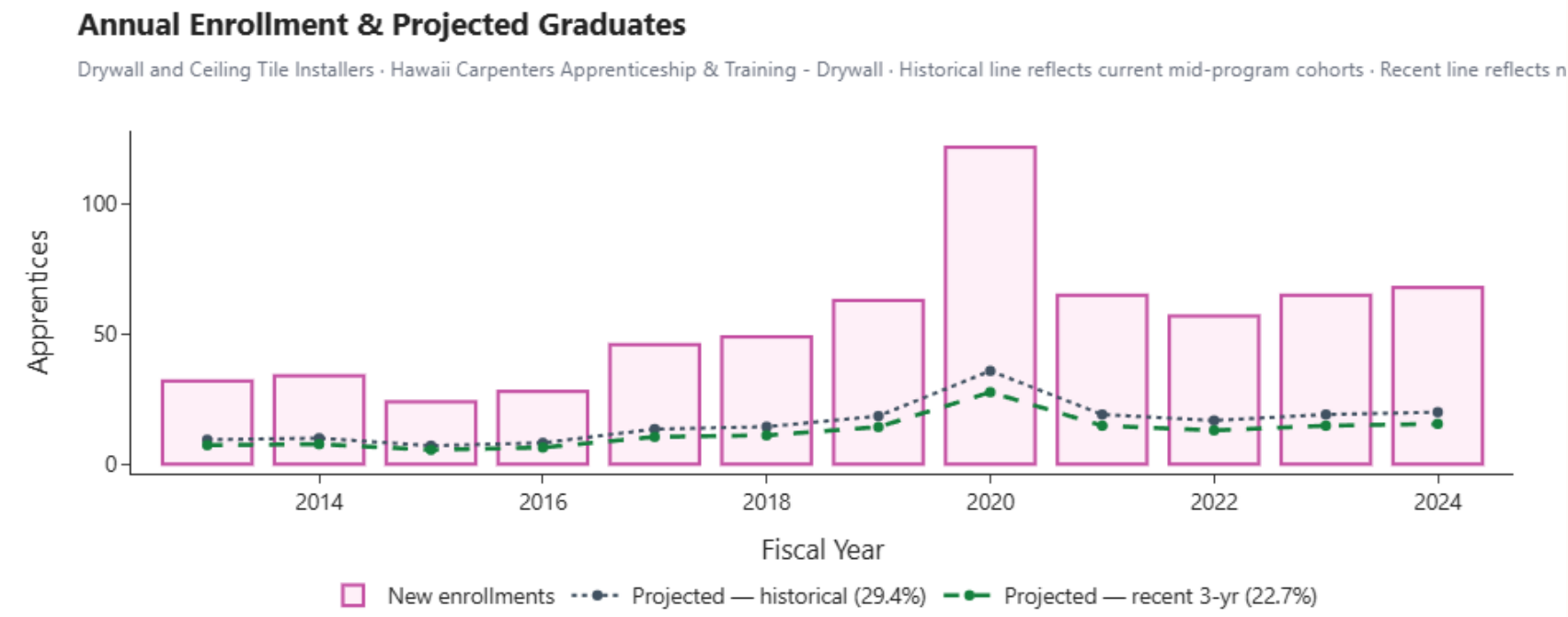
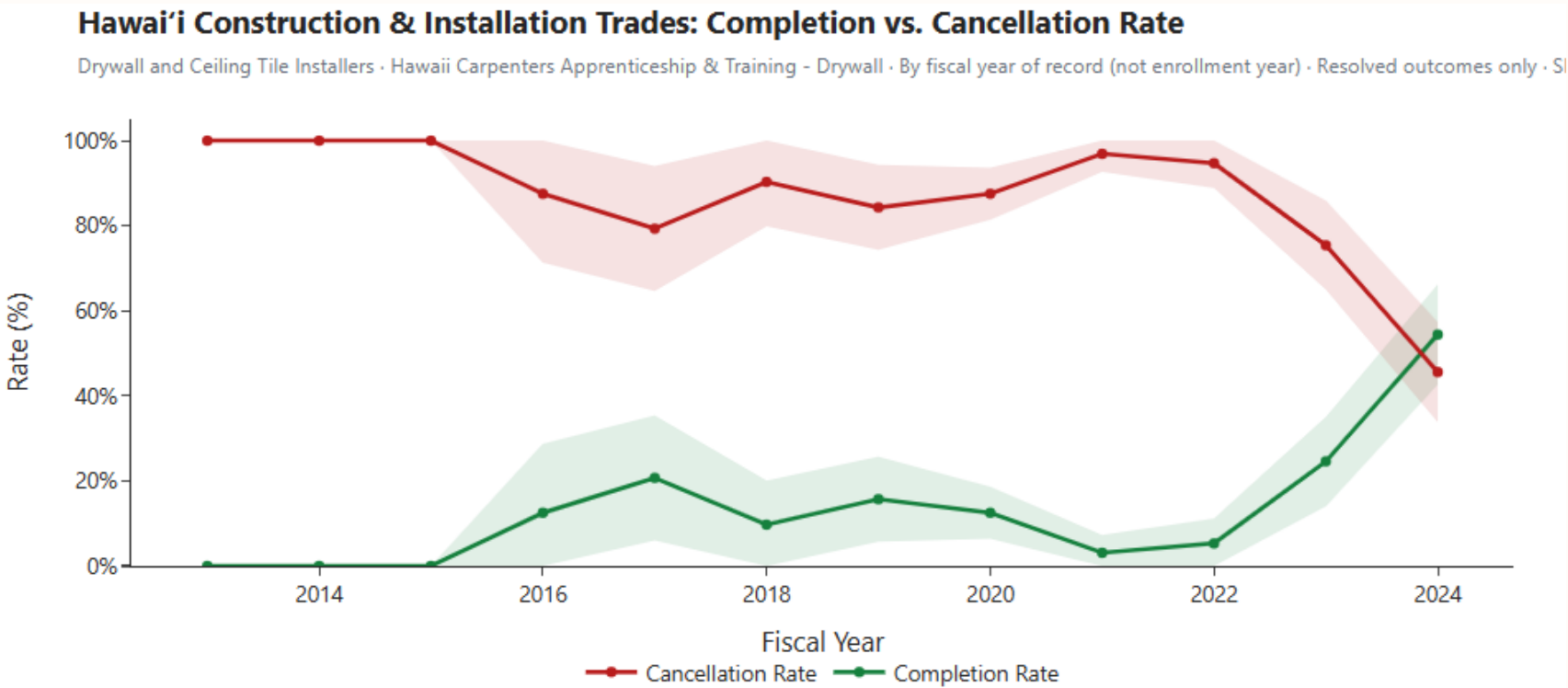
Drywall and Ceiling Tile Installers · Hawaii Carpenters Apprenticeship & Training - Drywall · n=633 · Expected completion: 4.0 years · Wage assumed at \$20/hr, 2080 hrs/yr — illustrative only



- 50% of cancellations 2021 and later happen ~20 months earlier than before 2021
- That's ~\$68k less invested per apprentice before they leave
- Caution: 2021+ cohort hasn't had time to mature; longer-term comparisons may still shift

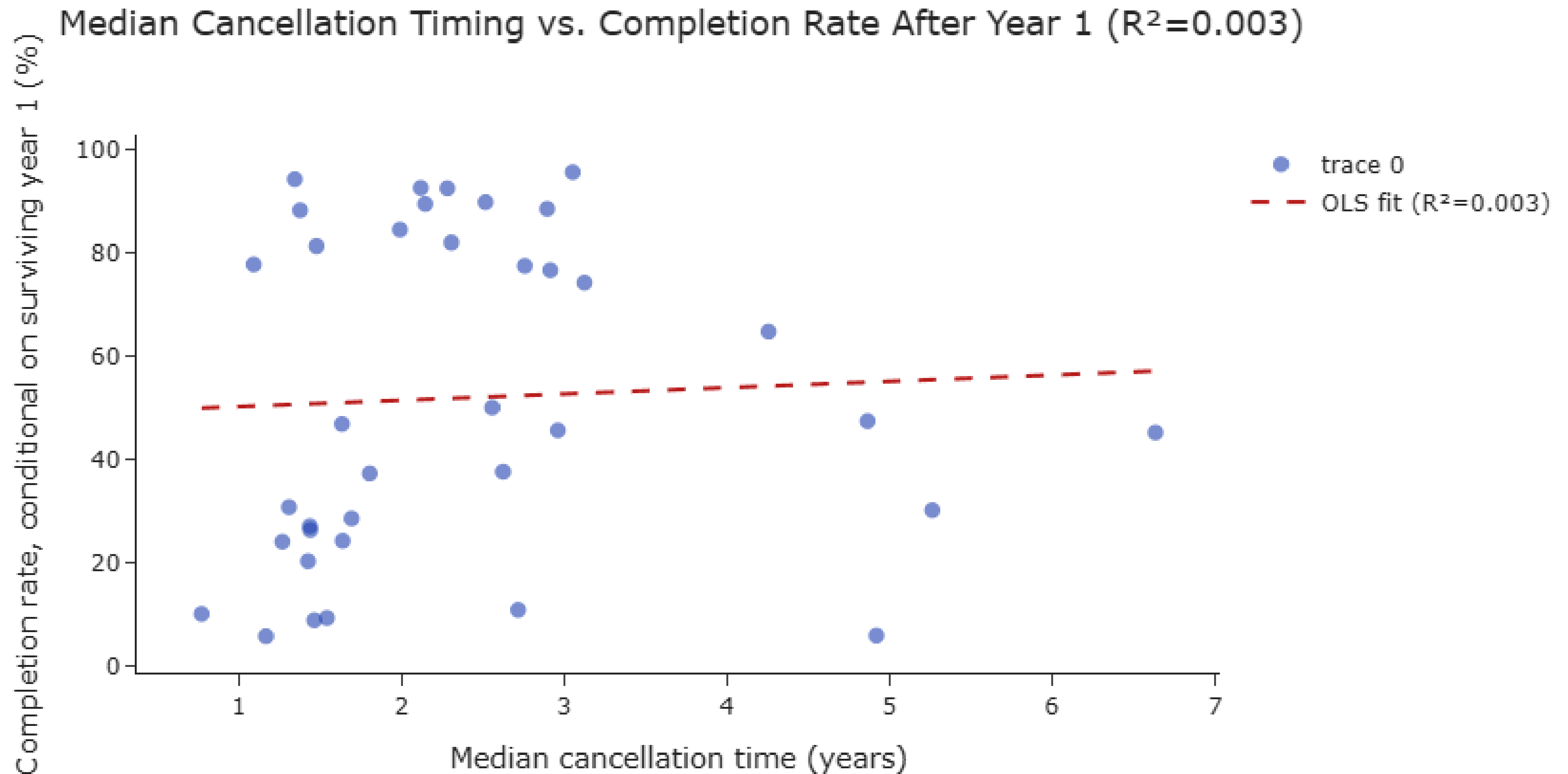
When do apprentices leave or finish?

Drywall and Ceiling Tile Installers · Hawaii Carpenters Apprenticeship & Training - Drywall · n=633 · Expected completion: 4.0 years · Wage assumed at \$20/hr, 2080 hrs/yr — illustrative only



Could this be a signal for success?

- Similar amount of enrolled
- Yet completion rate skyrocketed
- More savings from earlier exits could redistribute resources for existing apprentices



Hypothesis: Earlier cancellation %= Greater completion %
Most Plausible Answer: Perhaps not

EXECUTIVE RECOMMENDATION:

Early exits are a diagnostic signal, **not a KPI. (Simpson's Paradox)**

The real opportunity is reducing preventable trade exists, which are more costly and represent avoidable loss of talent.

Focus on:

- Identify the drivers of cancellations between Years 1-3, where most attrition occurs.
- Audit supervision and jobsite placement quality during the transition to higher responsibility.
- Benchmark against programs with stronger mid-program retention to identify practice worth adopting.

Do not optimize for earlier exits. The goal is better matching and support, not accelerating attrition.

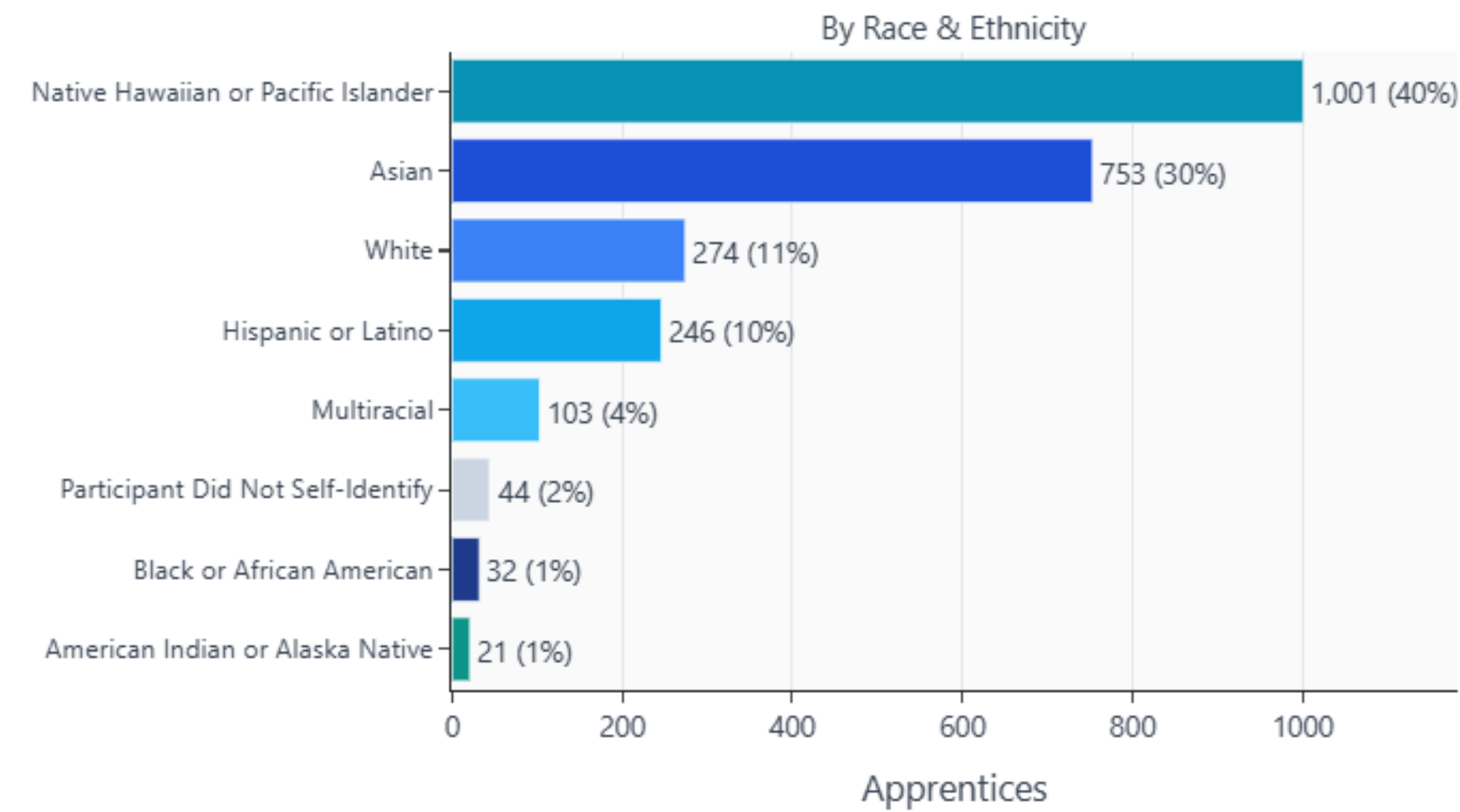
Note: Annual completion rates reflect outcomes from multiple older cohorts.

Survival curves isolate new cohorts only, which have not yet reached the completion window.'

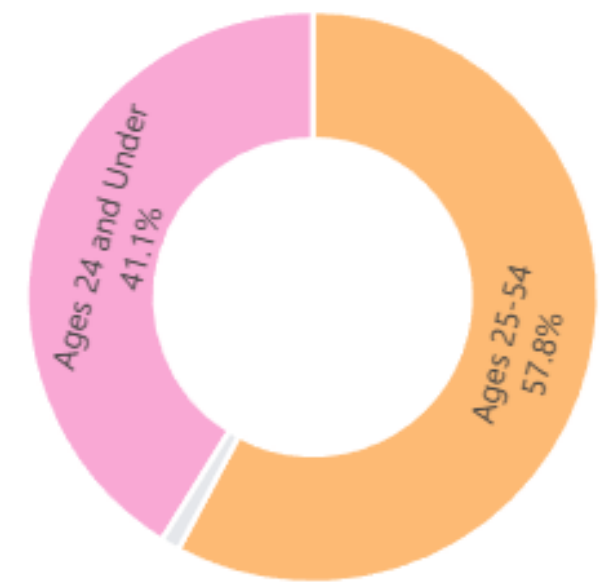


Demographics

Carpenters · Hawaii Carpenters Apprenticeship & Training · n=2,474 · categories below 10 grouped



By Age Cohort

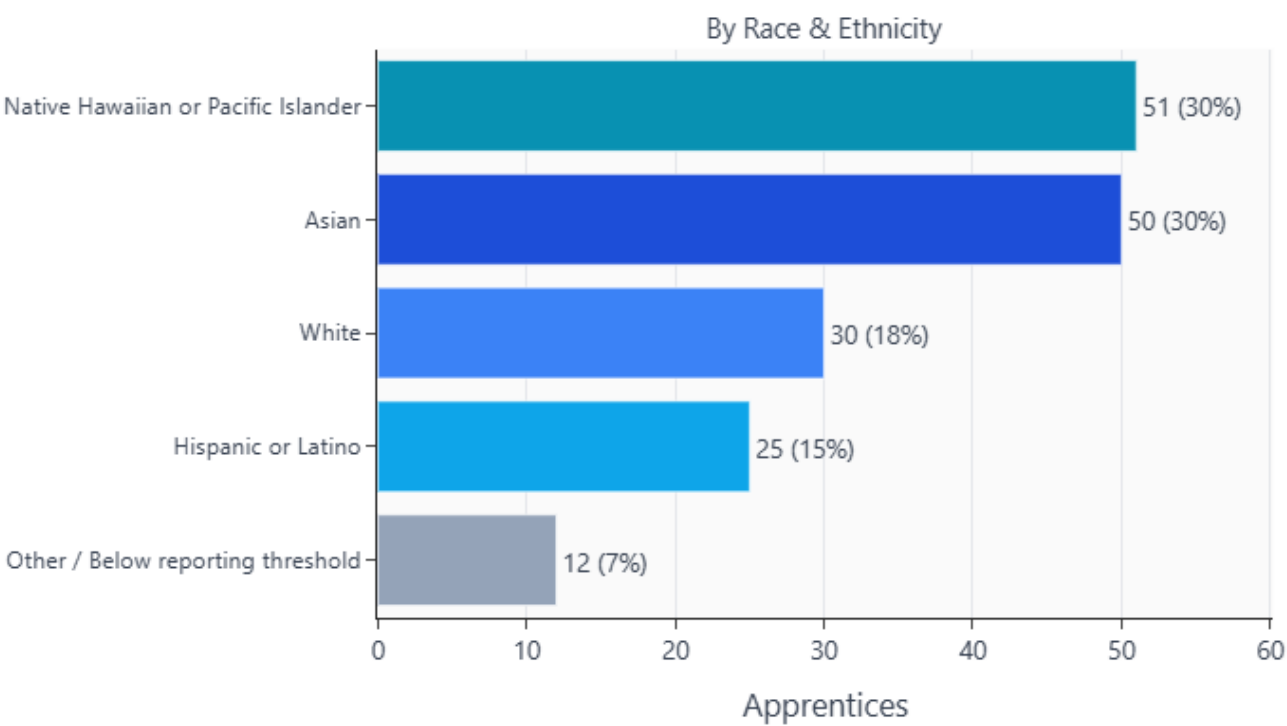


Demographics Facts:

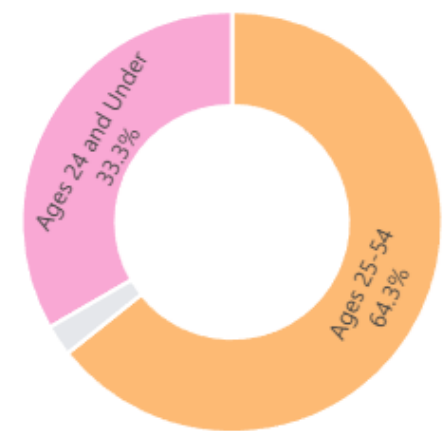
- Pacific Islanders make up roughly ~11% of state population (US BLS), yet disproportionately more likely to enter carpenter field

Demographics

Carpenters · ASSOCIATED BUILDERS & CONTRACTORS · n=168 · categories below 10 grouped



By Age Cohort



- Contrary to my own personal assumption, the majority of cohorts are older than 25

Areas of Improvement:

- Disaggregate it by island (e.g., Big Island, Maui) and add other islands (e.g., American Samoa, Guam, Puerto Rico)
- Identify greater socioeconomic refinement to identify casualty rather than association
- Tinker on how create more value for programs
- Upload dashboard as interactive website

My SCiPE CI-SIP Journey:

- An opportunity to an issue for Hawaii more deeply
- A chance to learn hands-on
- Expand the conversation beyond “just go into trade” for my state

“

**No effect like a problem exists
without a structure producing it.**

Anthony Zhong

Resources:

Apprenticeship Program. (n.d.) Hawaiian Electric,

www.hawaiianelectric.com/careers/apprenticeship-program.

Apprentice Program. (2023). Navy.mil.

<https://www.navsea.navy.mil/Home/Shipyards/PHNS-IMF/Careers/Apprenticeship/Apprenticeship-Postings/>

HireNet Hawaii - Employment Projections. (2026). Hirenethawaii.com.

<https://www.hirenethawaii.com/vosnet/gsipub/documentView.aspx?docid=729>

Office of Apprenticeship. (2024). *Apprentices by State Dashboard*. Apprenticeship.gov.

<https://www.apprenticeship.gov/data-and-statistics/apprentices-by-state-dashboard>

United States Census Bureau. (2024). *QuickFacts: Hawaii*. Census Bureau QuickFacts;

United States Census Bureau.

<https://www.census.gov/quickfacts/fact/table/HI/PST045224>



Visit my github repo!

<https://github.com/anthonyyzhong/sci-pe-fellowship-apprenticeship-code#>

Will soon publish interactive dashboard website for the public!

Mahalo to SC�PE CI-SIP cohort, Dr. Alex Stokes, Shivani Tankai, and

GROUP 4: The Bulls and the Bears

- Mentor: Marissa Halim
- Mentor: Aaron Mclean
 - Peer: Binh Giang
 - Peer: Wilson Chau

Mahalo for listening!



HAWAI'I DATA SCIENCE



NSF - SC�PE Cyberinfrastructure Pacific Professionals (CI-PP)

Award #2417946



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